MILITARY SPECIFICATION

INDICATOR, FREQUENCY CHANNEL
ID-2121A/ARC, ID-2229A/ARC, and ID-2303/ARC

This specification is approved for use within the Naval Air Systems Command, Department of the Navy and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the performance, design, and test requirements for the ID-2121A/ARC, ID-2229A/ARC and ID-2303/ARC Indicators, Frequency Channel hereinafter referred to as the indicator. The indicators covered in this specification are used with the receiver-transmitters of the ARC-182 family.

1.2 Classification. Three types of indicators are covered by this specification. The principle difference between the three indicators is the colors of the panel lighting. The indicators and respective lighting are as follows:

a.	ID-2121A/ARC	Red
b.	ID-2229A/ARC	White
c.	ID-2303/ARC	Blue-White

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Air Engineering Center, Systems Engineering and Standardization Department (SESD), Code 53, Lakehurst, NJ 08733-5100 by using the self-addressed Standardization Document Improvement or by letter.

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

Federal

QQ-P-416	Plating, Cadmium (Electrodeposited)

Military

MIL-B-5087	Bonding, Electrical, and Lightning Protection, for Aerospace Systems
MIL-W-5088	Wiring; Aircraft, Selection and Installation of
MIL-E-5400	Electronic Equipment, Aircraft, General Specification for
MIL-T-5422	Testing, Environmental, Aircraft Electronic Equipment
MIL-C-6781	Control Panel; Aircraft Equipment, Rack or Console Mounted
MIL-P-7788	Panel, Information, Integrally Illuminated
MIL-E-17555	Electronic and Electrical Equipment, Accessories, and Repair Parts, Packaging and Packing of
MIL-T-18303	Test Procedures; Preproduction, Acceptance and Life for Aircraft Electronic Equipment, Format for
MIL-N-18307	Nomenclature and Identification for Electronic, Aeronautical, and Aeronautical Support Equipment Including Ground Support Equipment
MIL-C-25050	Colors, Aeronautic Lights and Lighting Equipment, General Requirements for
MIL-M-38510	Microcircuits, General Specification for

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MIL-C-38999	Connectors, Electrical, Circular, Miniature, High Density, Quick Disconnect, Environmental Resistant, Removable Crimp Contacts; General Specification for
MIL-H-46855	Human Engineering Requirements for Military Systems Equipment and Facilities
MIL-P-55110	Printed Wiring Boards
MIL-C-85666	Control, Radio Set C-10319A/ARC, C-10776A/ARC, and C-11131/ARC

STANDARDS

Federal

FED-STD-595	Federal Standard Colors

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-275	Printed Wiring for Electronic Equipment
MIL-STD-415	Test Provisions for Electronic Systems and Associated Equipment, Design Criteria for
MIL-STD-454	Electronic Equipments, General Requirements for
MIL-STD-461	Electromagnetic Interference Characteristics Requirements for Equipment
MIL-STD-462 with Notice 2	Electromagnetic Interference Characteristics, Measurement of
MIL-STD-471	Maintainability Demonstration
MIL-STD-704	Electric Power, Aircraft, Characteristics and Utilization of
MIL-STD-781	Reliability Tests; Exponential Distribution
MIL-STD-785	Reliability Program for Systems and Equipment Development and Production

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MIL-STD-810	Environmental Test Methods
MIL-STD-883	Test Methods and Procedures for Microelectronics
MIL-STD-965	Parts Control Program
MIL-STD-1472	Human Engineering Design Criteria for Military System Equipment and Facilities
MIL-STD-1562	Lists of Standard Microcircuits
MIL-STD-2074	Failure Classification for Reliability Testing
MIL-STD-2076	Unit Under Test Compatibility with Automatic Test Equipment, General Requirements for
MIL-STD-2084	Maintainability of Avionics and Electronic Systems and Equipment, General Requirements for

2.1.2 Other Government documents. The following other Government documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DOCUMENTS

Naval Air Systems Command

WS-6536	Process Specification Procedures and Requirements for Preparations and Soldering of Electrical Connections

Department of the Navy

NAVMAT P-4855.1	Navy Power Supply Reliability
NAVMAT P-9492	Navy Manufacturing Screening Program

(Copies of specifications, standards and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Item description. The ID-2121A/ARC, ID-2229A/ARC and ID-2303/ARC Frequency Channel Indicators shall be self-contained units for cockpit mounting. The indicators shall contain circuits necessary to operate with a host receiver-transmitter and radio set control.

3.2 First article. When specified in the contract or purchase order (see 6.2.1), a sample shall be subjected to first article inspection (see 4.4 and 6.3).

3.3 Parts. In the selection of parts, the prime considerations are fulfillment of performance requirements, logistics supportability and reliability assurance in a cost effective design. The following shall govern in parts selection.

a. Subassemblies shall be designed and fabricated to be repairable. Maximum economic standardization of parts and materials shall be exercised. Parts selection shall be made in the following order of precedence:

(1) Military Standard parts shall be selected in accordance with MIL-STD-454, Requirement 22.

(2) Nonstandard parts previously qualified for other programs.

(3) Nonstandard parts.

b. Nonstandard parts and nonestablished reliability military standard parts shall be screened to the requirements of the most similiar military standard or standard nonestablished reliability part. Integrated circuits and hybrids shall use MIL-STD-883, Method 5004, Class B, as a basis for tailored screens.

3.3.1 Component rescreening. The contractor shall conduct 100 percent incoming inspection of all active components used in production equipment and spares. Functional and parametric checks of active components shall be performed at the components rated temperature extremes and at ambient. Components which are not compliant with their specification shall not be used in the ARC.

3.3.2 Parts derating and application. All parts used shall be applied well within their ratings. The derating shall encompass the appropriate and meaningful application conditions such as voltage, current, power, temperature, mechanical, and duty cycle. Electronic and electromechanical parts shall conform to Table I electronic parts derating for worst case electrical and environmental stress unless formal written approval is received from the procuring activity prior to incorporation into the design. Part level stress analysis shall be used to verify that all parameter stresses are within the derated values at worst case circuit and environmental conditions. The following limitations on parts usage shall apply:

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Table I. Electronic parts derating.

Part Type	Parameter	Maximum Allowable Application Stress
Capacitor	Applied Voltage (Ripple Voltage for Tantalum)	50% of rating 1/
Resistors	Power Voltage	50% of rating 70% of rating
Diodes	Power Current Voltage Junction Temperature	50% of rating 75% of rating 60% of rating 2/
Transistors	Power Voltage Junction Temperature	55% of rating 60% of rating 2/
Integrated Circuits		
Digital	Fan-Out, Fan-In Junction Temperature	80% of rating 2/
Linear	Power Junction Temperature	75% of rating 2/
Transformers, Inductive Devices	Parts shall be used at a winding temperature of at least 30 degrees below rated temperature. In addition, the maximum voltage between windings or between a winding and a shield shall be not greater than 60% of rated voltage for the steady state conditions and 90% of rated voltage with transients. General purpose inductors, audio and power transformers shall be applied at no more than 70% of rated current.	

1/ For solid tantalum and all electrolytic capacitors, the maximum applied voltage shall be not greater than 70 percent of the vendor rated value. "Applied voltage" is the sum of the applied peak ripple voltage and the applied DC voltage.

2/ Semiconductor devices shall be mounted and operated in such manner to insure worst-case junction temperature is not greater than 110 degrees C at +71 degrees C ambient temperature. Any deviation requires procuring activity approval. Power semiconductor devices, on a very selective basis, may be mounted and operated with worst case junction temperatures not greater than 150 degrees C at +71 degrees C ambient temperatures.

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a. Prohibited application

(1) Non-hermetically sealed semiconductors except power transistors
(2) Photo couplers (photo transistors)
(3) Hot carrier (Schottky) rectifiers
(4) Microcircuit sockets
(5) Non-hermetically sealed wet tantalum capacitors

3.3.3 Parts control program. The contractor shall establish and maintain a parts control and standardization program in accordance with the requirements of MIL-STD-965, Procedure I and as specified herein. The program shall control part standardization and the criteria for the selection, application, and testing of parts to achieve the reliability required. As a minimum, a parts program equivalent to that specified herein shall be required of subcontractors and suppliers.

3.3.4 Non-standard parts approval. The contractor shall obtain procuring activity approval of all non-standard parts used in the equipment. Standard parts are defined in MIL-E-5400. In addition to the MIL-E-5400 requirements for standard parts, microelectronic parts are standard only if:

a. The device is listed in MIL-STD-1562, and

b. Sources (at least two) are listed on the Qualified Parts List (QPL), and

c. The part is procured from one of the qualified sources, and

d. Quality level "B" (or better) of MIL-M-38510 is required.

The approval procedure shall be in accordance with MIL-E-5400 and MIL-STD-965, Procedure I. In addition to the requirements of MIL-E-5400, approval of non-standard parts will be based on:

a. Suitability for the application

b. Conformance to this specification and to MIL-E-5400

c. Interchangeability

d. Screening (burn-in) method

3.3.5 Samples of non-standard parts. Samples of non-standard parts are not required unless specified in the contract. Where non-standard parts samples are specified in the contract, the sample shall be in accordance with MIL-E-5400 (see 6.2.1).

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3.3.6 Cabling and connections.

3.3.6.1 Cables and connectors. The indicator shall provide for the use of cables and connectors in accordance with MIL-E-5400. The design of the connectors shall be such that improper assembly, mating or installation is impossible. Non-RF connectors shall meet the requirements of MIL-C-38999. All external connectors shall be scoop-proof, have cadmium plating finish in accordance with QQ-P-416 (over a suitable underplating) to withstand a 500 hour salt test, and provide 360 degrees circumferential grounding techniques for grounding prior to contact engagement.

3.3.6.2 Interconnection cabling. The indicator shall be capable of required operation using external wiring in accordance with the applicable requirements of MIL-W-5088. The external wiring shall be unshielded, except that a minimum number of the individual wires may be shielded when demonstrated as necessary to meet interference control requirements and provide the assembly of the cable to its plugs may be easily accomplished. External cables and that portion of the connectors attached to the cables shall not be supplied as part of the indicator.

3.3.7 Control panels. The control panels shall conform to the applicable requirements of MIL-C-6781.

3.3.7.1 Lighted panels - console mounted equipment. The plastic panels for all console mounted control panels, indicators, and radio sets shall conform to MIL-P-7788, Class 1-R, 1-W, or 1-BW Type V, with both 5 and 28 volt (ac and dc) lamp systems installed. Separate input power pins shall be provided for the two lighting systems. Color of panel lighting shall be in accordance with MIL-C-25050, as selected from the options listed in Table II by nomenclature assignment, or by description on purchase orders.

Table II. Panel lighting options

Nomenclature (Revisable)	Color (Short Name)	Color per MIL-C-25050	MIL-P-7788 Class
ID-2121A/ARC	IPL RED	Type 1 (f)	1-R
ID-2229A/ARC	IPL WHITE	Type 1 (g)	1-W
ID-2303/ARC	BLUE FILTERED WHITE	Type 1 (h)	1-BW

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3.3.7.2 Console mounted equipment function marking. Console mounted control panels shall be permanently and clearly marked to indicate the function of the equipment with which they are associated. Console mounted control panels shall be identified by the abbreviation VHF/UHF located on the front panel.

3.3.7.3 Control knobs and functions. The number of controls required to operate the indicator shall be limited to and consistent with performing all the functions required by this specification. Closely associated controls such as control knobs on concentric shafts, shall be easily identified by contrasting colors of black and gray.

3.3.8 Interchangeability. Physical and functional interchangeability shall exist for all units and replaceable assemblies, subassemblies, and parts of the indicator in accordance with MIL-E-5400. The indicator shall be constructed to preclude the requirement for any adjustment when replacing an item by another item of the same type designation.

3.4 Design and construction. The indicator shall conform with all the applicable requirements of MIL-E-5400 for design and construction except where otherwise specified herein. In any case, it is a condition of final acceptance that the indicator shall meet all the performance requirements of this specification.

3.4.1 Reliability. When specified in the contract or purchase order, the contractor shall establish a reliability program conforming to MIL-STD-785 (see 6.2.2). Task 104 of MIL-STD-785 shall be implemented to establish and maintain a closed-loop failure reporting system. This system will identify failure trends and patterns at all levels of assembly prior to acceptance of the hardware and will also identify failure trends related to field failures throughout the contract period. The contractor shall develop and implement corrective action for significant failure trends (see 6.2.2). The reliability program shall include provisions for assuring the reliability of the indicator or any portion of the equipment when supplied for use as spare or repair part.

3.4.1.1 Operational stability. The indicator shall operate with specified performance for at least the specified mean (operating) time between failures without the necessity for readjustment of any controls which are inaccessible to the operator during normal use.

3.4.1.2 Operating life. The indicator shall have a total operating life of 20,000 hours with minimal servicing and replacement of parts. Parts requiring scheduled replacement shall be specified by the contractor.

3.4.1.3 Reliability in MTBF (mean time between failure). The indicator, including, any built-in-test (BIT) provisions, shall have a specified mean (operating) time between failures, upper test MTBF (O_O) of 5,000 hours (as specified in MIL-STD-781), and a lower test MTBF (O_1) of 2,500 hours when tested and accepted as specified under the requirements of 4.4.5 and 4.5.4.

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3.4.2 Maintainability. Maintainability (built-in test features, construction and packaging, provision for test points and other maintainability parameters) shall be as specified in MIL-STD-2084.

3.4.2.1 Compatibility with ATE. The indicator shall be compatible with Automatic Test Equipment (ATE) for WRA testing and fault isolation to a faulty SRA in accordance with MIL-STD-2076.

3.4.2.2 Maintainability design. All electrical or electronic circuits and parts shall be packaged on replaceable and repairable, plug-in modules. Circuit breakers, indicators, and other items which may logically be bolted and soldered, or otherwise more rigidly electrically and mechanically fastened to the indicator housing shall not be considered modules. SRAs (Shop Replacement Assemblies) shall be ORAs (Quick Replaceable Assemblies) as specified in MIL-STD-2076. SRA arrangement shall be such that access to any SRA does not require the removal of adjacent SRAs or parts other than access panels. The use of a special tool or tools shall not be required for SRA removal. Fuses may be used in lieu of circuit breakers.

3.4.2.2.1 Keying. Connectors shall be keyed to prevent the insertion of a connector or subassembly into an improper location within or on the indicator. All connectors shall be legibly marked with reference designations.

3.4.2.2.2 Access panels. Access panels shall be retained by fasteners suitable to the procuring activity.

3.4.2.2.3 Service access. Fasteners required to be loosened for removal of the WRA shall be held by retaining devices suitable to the procuring activity.

3.4.2.2.4 Adjustments. No adjustment or alignment shall be required at the organizational level. Adjustment or alignment if required at a higher maintenance level (intermediate or depot), shall not require the removal of the part to be adjusted or aligned, or of the hardware element to which it is attached.

3.4.2.2.5 Test points. All test points shall be in accordance with MIL-STD-2084 and MIL-STD-415 to accommodate approved test equipment. MIL-STD-2084 takes precedence over MIL-STD-415 requirements. The sub-SRA test points and related connector requirements of MIL-STD-2084 are waived where prohibited by space or circuit performance considerations. The sub-SRA test points and related connector requirements philosophy shall be reflected in appropriate maintenance data. Conformal coating may be applied to the PC pads used as test points. Injection test points in the BIT circuits are not required.

3.4.2.2.6 Organizational level maintainability. The indicator shall provide the operator or organizational maintenance technician, while airborne or on the ground with performance or readiness test and fault (GO/NO-GO) indication.

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3.4.2.2.6.1 Performance or readiness test requirements. BIT provisions shall be incorporated in the indicator to provide a GO/NO-GO indication of indicator readiness. These provisions may be in conjunction with, but not necessarily as part of, the fault isolation features. The operation of the performance or readiness test provision shall be both a manually energized or continuous test mode of operation. RF radiation is not permitted during performance readiness tests. Normal equipment performance may be interrupted during the manually initiated BIT sequence and, if necessary, switches may be used in series paths for purposes of introducing BIT stimuli or measuring equipment performance. Self-test of the BIT is not required.

3.4.2.2.6.2 MTTR (mean time to repair). For organizational maintenance, the MTTR shall be not greater than 15 minutes to remove and replace and verify operation of the replaced equipment. The indicator design shall be such that no ancillary test equipment is required for organizational level maintenance.

3.4.2.2.7 Intermediate level maintainability. Intermediate level maintenance may be performed with the assistance of separate test equipment to isolate a fault to the SRA level. The MTTR to isolate the fault to a SRA, replace the SRA, and return to full performance at the intermediate level maintenance shall be not greater than 20 minutes.

3.4.3 Transportability. The indicator shall be transportable by air, rail, truck and ship when packaged as specified in Section 5.

3.4.4 Standard conditions. The following conditions shall be used to establish normal performance characteristics under standard conditions and for making laboratory bench tests, except that required field tests may be conducted under outside ambient conditions:

a.	Temperature	Room ambient (25 $\pm$ 10°C)
b.	Altitude	Normal ground
c.	Vibration	None
d.	Humidity	Room ambient up to 90 percent relative humidity
e.	Input power voltage	28 $\pm$ 0.5 VDC

3.4.4.1 Service conditions. The indicator shall operate as required under any of the environmental service conditions or combination of these conditions as specified in MIL-E-5400 for Class 2 equipment, except as modified herein.

3.4.4.1.1 Sinusoidal vibration. The indicator shall not require isolators or rear support and shall operate as required when subjected to sinusoidal vibration in accordance with Test Procedures of MIL-T-5422 to the requirements of Curve IIIB of MIL-E-5400.

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3.4.4.1.2 Random vibration. The indicators shall operate as required when subjected to the random vibration requirements specified in 4.4.4.

3.4.4.1.3 Rain protection of console mounted equipment. The portion of the console mounted controls which are normally exposed when the equipment is mounted in a console shall meet the rain requirements of MIL-STD-810, Method 506.1.

3.4.4.1.4 Thermal conditions. The indicators shall operate as required under the thermal conditions specified by MIL-T-5400, Class 2, without the requirement for auxiliary cooling.

3.4.4.1.5 Temperature shock. The indicators shall meet the temperature shock requirements of MIL-T-5422.

3.4.4.2 Warm-up time. The time required for the indicators to warm up prior to operation shall be kept to a minimum and shall be not greater than 10 seconds at standard conditions. Under service conditions, the warm-up time shall be not greater than 20 seconds.

3.4.4.3 Electromagnetic interference (EMI). The indicators shall comply with the requirements MIL-STD-461 with Notice 3 for Class 1/A equipment except as modified in Tables III, IV and herein. These requirements shall be met under all combinations of operating modes and frequencies of the indicator.

3.4.4.3.1 Conducted and radiated emissions limits. The conducted and radiated emissions of the indicators shall comply with the limits of the following requirements of MIL-STD-461:

a. CE01

b. CE02

c. CE03

d. CE04

e. RE01

f. RE02

3.4.4.3.2 Conducted and radiated susceptibility limits. The indicator shall comply with conducted and radiated susceptibility limits of the following requirements of MIL-STD-461:

a. CS01

b. CS02

c. CS06

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d. RS01

e. RS02

f. RS03

The indicator shall exhibit no susceptibility as specified in 3.4.4.3.3 when subjected to an RS03 test to the limits of Table III. In addition, the indicator shall exhibit no susceptibility in accordance with the criteria of 3.4.4.3.4b when subjected to signals at the levels of Table IV.

3.4.4.3.3 Susceptibility. The following criteria shall be applied to determine whether the indicator is susceptible:

a. Any failure of the indicator to comply with the requirements of 3.6.6 shall be considered an indication of susceptibility.

b. Any repeatable non-command change in the modes of operation or present frequencies of the indicator shall be considered an indication of susceptibility. This shall apply whether the failure is transient (momentary) or non-transient (permanent) in nature. In addition the inability to change radio frequency or mode of operation while exposed to the susceptibility signal shall be considered an indication of susceptibility.

3.4.4.3.4 Ground plane interference. All indicator interface circuits shall be capable of specified performance when subjected to the following aircraft chassis noise between the WRA's.

a. Three volts rms from 320 Hz to 500 Hz. (Not to be greater than 150ma rms applied current.)

b. One volt rms from 500 Hz to 20 MHz. (Not greater than 150ma rms applied current for frequencies between 500 Hz and 50 kHz. Not to be greater than 1 watt (CW) from a 50-ohm source for frequencies above 50 kHz.)

c. ± eight volt pulses, 70 usec wide at 100 pps. (Not to be greater than 15a peak current spikes.)

In addition to the susceptibility definitions of 3.4.4.3.3, the radio shall maintain intelligible communications during a "talk-thru" test with another radio set using a low level signal (not to exceed 20 microvolts input to the receiver). The test will be performed with a test set up employing a RT-1250A/ARC, RT-1324A/ARC, or equal receiver-transmitter.

3.4.4.3.5 Conducted susceptibility test, interconnecting and signal leads. The indicator shall exhibit no susceptibility, as specified by 3.4.4.3.3 when subjected to 100 MV rms 2 to 30 MHz coupled into its interconnecting and signal leads.

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Table III. RS03 signals

Frequency (MHz)	Modulation				Power Density (mW/cm^2) Peak 5/	Avg	v/m (ref) 6/
0.014-850	AM90%	1 kHz Sine Wave			-		20
215-225	PAM	20 usec	PW	200 pps4/	.317	-	35
215-225	PAM	200 usec	PW	200 pps	.317	-	35
400-450	PAM	15 usec	PW	300 pps	.106	-	20
850-940	PAM	10 usec	PW	1000 pps	.106	-	20
850-940	PAM	125 usec	PW	300 pps	.106	-	20
1215-1365	PAM	10 usec	PW	3000 pps	.106	-	20
1365-2900	PAM	1 usec	PW	1000 pps	.106	-	20
2900-3100	PAM	3 usec	PW	1000 pps	1.6	-	78
2900-3100	PAM	70 usec	PW	200 pps	1.6	-	78
3100-9000	PAM	1 usec	PW	1000 pps	.106	-	20
9000-14000	PAM	0.2 usec	PW	1000 pps	.106	-	20
14000-18000	PAM	0.2 usec	PW	50000 pps	.106	-	20

1/ Pulse Amplitude Modulation.
2/ Microseconds.
3/ Pulse Width.
4/ Pulses Per Second.
5/ Peak Power Density for PAM signals is specified as the power density measured by an average power measuring device if the generator is run CW; average power density within the pulse envelope.
6/ Volts Per Meter equivalent shown for reference only.
7/ Frequency ranges listed with more than one modulation shall be repeated as necessary to include all different modulations and field strength shown.

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TABLE IV. High level RS03 signals

Frequency (MHz)	Modulation				Power Density (mw/cm^2) 5/	v/m (ref) 6/	Scan Rate (sec) 7/	Dwell Time (msec) 7/
220 8/	PAM1/	10 usec2/	PW3/	200 pps4/	10.6	200	4	500
220	PAM	200 usec	PW	200 pps	10.6	200	4	500
425	PAM	15 usec	PW	300 pps	10.6	200	10	200
900	PAM	2 usec	PW	1000 pps	10.6	200	5	50
900	PAM	125 usec	PW	300 pps	10.6	200	5	50
1250	PAM	10 usec	PW	3000 pps	10.6	200	5	50
1350	PAM	5 usec	PW	3000 pps	10.6	200	5	50
3000	PAM	3 usec	PW	1000 pps	10.6	200	5	50
3000	PAM	70 usec	PW	200 pps	10.6	200	5	50
3600	PAM	1 usec	PW	1000 pps	10.6	200	5	50
5600	PAM	1 usec	PW	600 pps	10.6	200	5	50
9100	PAM	0.5 usec	PW	1000 pps	10.6	200	5	50
9375	PAM	0.2 usec	PW	1000 pps	10.6	200	5	50
13900	PAM	0.2 usec	PW	50000 pps	10.6	200	0	CONT
15500	PAM	0.2 usec	PW	50000 pps	10.6	200	0	CONT
33200	PAM	0.2 usec	PW	2000 pps	10.6	200	0	CONT

1/ Pulse Amplitude Modulation.
2/ Microseconds.
3/ Pulse Width.
4/ Pulses Per Second.
5/ Peak Power Density for PAM signals is defined as the power density measured by an average power measuring device if the generator is run CW; average power density within the pulse envelope.
6/ Volts Per Meter equivalent shown for reference only.
7/ Scan and Dwell refer to simulated antenna rotation. The signal shall be disabled except for the specified dwell. The signal shall be periodically enabled for the specified dwell time with the period equal to the specified scan time.
8/ Frequency ranges listed with more than one modulation shall be repeated as necessary to include all different modulations shown.

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3.4.4.3.6 Grounding requirements. The grounding system between WRA's of the equipment shall be compatible and consistent with the aircraft grounding scheme specified in 3.4.4.3.6.1 through 3.4.4.3.6.3. All WRA's shall be grounded in such a manner as to prevent ground loops and common ground returns for signal and power circuits, provide effective shielding for signal circuits, minimize EMI, and to protect personnel from electrical hazards.

3.4.4.3.6.1 Chassis grounds. A ground wire of minimum length connected internally to the WRA chassis shall be provided at a pin on each primary power connector. No circuit shall be allowed to utilize this wire as its primary return.

3.4.4.3.6.2 Shielding grounding. A separate connector pin shall be provided for each aircraft wire shield unless suitable grounding is provided in external cable backshells. The connector shield pins shall be grounded to the equipment chassis inside the WRA adjacent to the connector mounting and by the shortest means practical.

3.4.4.3.6.3 Component grounds. All externally exposed metal parts, shields, control shafts, switch handles, connectors, and bushings shall be grounded to the chassis.

3.4.4.3.7 Shielding gaskets. Shielding gaskets and similar devices for RF applications shall avoid the use of materials which represent severe dissimilar metal combinations with the interface.

3.4.5 Dimensions. The dimensions of the indicator shall be as shown on Figure 1.

3.4.6 Weight. The weight of the indicator shall be not greater than 1.1 pounds.

3.4.7 Color. The exterior color of the indicator shall be lusterless black in accordance with FED-STD-595, Color No. 37038.

3.4.8 Nameplates and identification marking. Nameplate approval and identification marking shall be in accordance with MIL-N-18307.

3.4.8.1 Decals, nameplates, and marking locations. All decals, nameplates, and markings shall be affixed to the indicator as shown on Figure 1.

3.4.8.2 Serial numbers. Serial numbers will be assigned by the procuring activity for the indicator under contract or purchase order. The contractor shall serialize sequentially all indicators which are delivered under the contract.

3.4.9 Input electrical power.

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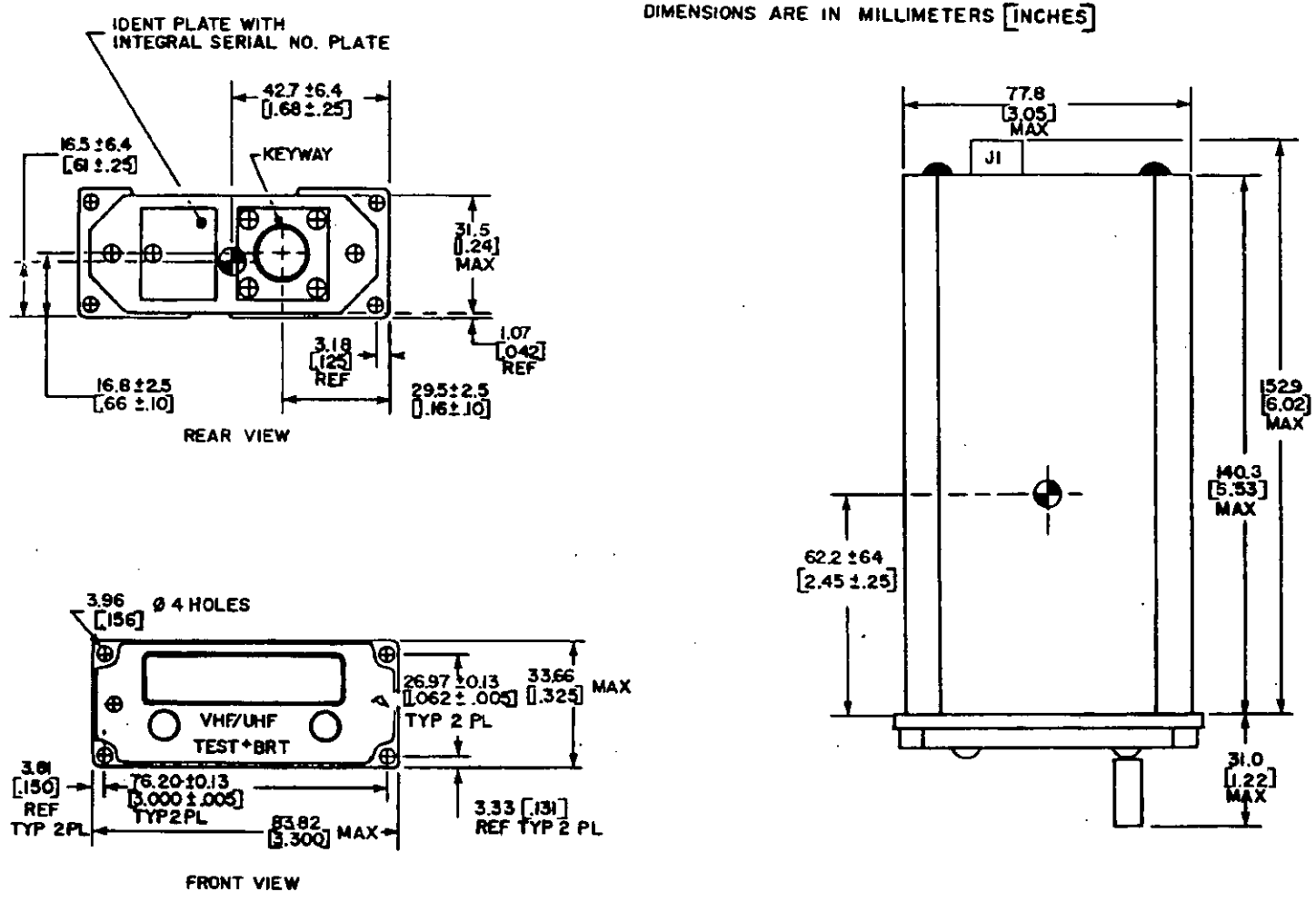


FIGURE 1. Outline dimensions of the indicators

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3.4.9.1 Operating power. The indicator shall meet all applicable requirements of MIL-STD-704 and shall give specified performance when energized from the following power sources having characteristics and limits as specified in MIL-STD-704. The voltage supplied shall be 28 volts DC. The required power shall be not greater than 15 watts. The operating power requirements shall not include the power requirements of 3.4.9.2.

3.4.9.2 Lighting power. Input power for lighting of any lighted panel shall not require more than 5.0 watts at 5 or 28 volts (ac or dc).

3.4.9.3 Degraded performance. Degraded performance will be permitted for voltage transients with durations of 75 microseconds to 0.5 second during normal electrical system operation. Operation shall return to normal within five seconds with no resulting damage to the indicator.

3.3.9.4 Emergency voltage. During operation in the emergency steady state electric power condition as specified in MIL-STD-704, the equipment shall remain safe and frequency readouts and shall be maintained. The emergency power source will impose an interval of zero voltage at time on loss of primary power for a maximum period of 30 seconds, but shall return to performance as specified above after return to emergency power conditions in accordance with MIL-STD-704.

3.4.9.5 Undervoltage protection. The indicator shall not be damaged by voltages below the minimums specified by MIL-STD-704, and shall be capable of resuming normal operation when the voltage returns within normal limits.

3.4.9.6 Reverse polarity protection. The indicator shall be protected such that it will not be damaged by application of reverse polarity voltages.

3.4.10 Manufacturing requirements. The indicator shall be manufactured in accordance with manufacturing processes, workmanship, and quality assurance provisions of MIL-STD-454, MIL-STD-275, MIL-P-55110, WS-6536, NAVMAT P-4855.1, and NAVMAT P-9492.

3.5 Performance. Unless otherwise specified herein, values set forth to establish the specified performance requirements apply to performance under both standard and extreme service and input power conditions. When reduced performance under the extreme conditions is acceptable, tolerances or values setting forth acceptable variations from the performance under the standard conditions will be specified.

3.5.1 Operation. The indicator shall provide the functions and capabilities listed in section 1 for operation in military aircraft and helicopters. Additionally the indicator shall be compatible with associated equipment listed in 6.5 to provide operations specified in 3.6.

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3.6 Detail requirements.

3.6.1 Function. The indicator shall provide remote presentation of the frequency, channel, operational or test status of the AN/ARC-182(V) control unit or receiver-transmitter readout.

3.6.2 Contents. The indicator shall be one self-contained unit suitable for cockpit mounting, and shall contain circuits and elements for the following capability.

a. Six digit readout of frequency, channel or status display.

b. Power supply circuitry for operation from 28 volts dc primary power.

c. Conversion circuitry for readout presentation from serial digital information as specified in MIL-C-85666.

d. Circuitry for readout illumination control and digital lamp test.

3.6.3 Controls. The front panel controls shall consist of a dimmer control and lamp test switch. The dimmer control shall vary the illumination intensity of the frequency/channel display. Activation of the lamp test switch shall illuminate all the segments in all six digital displays.

3.6.4 Frequency and channel display. An incandescent or equal six digit frequency display shall be provided on the front of the indicator to display the operating radio frequency. A decimal shall also be lighted to identify the MHz position of three significant digits (i.e., 225.000). For preset channel number display, only one or two digits (for preset channel number 1 through 28) shall be lighted with the remaining digits unlighted. The display shall be arranged horizontally in a centrally located position on the front panel of the indicator. The displays shall be such that the digits can be easily read from a distance of 2.0 feet at any angle up to at least 30 degrees from the display centerline perpendicular to the surface of the front panel. Provisions shall be made for easy removal and replacement of the display digits without disturbing the operating functions or removing the controls.

3.6.5 Readout presentation. The indicator shall normally be operated from a panel mounted receiver-transmitter RT-1324A/ARC or equal or a C-10319A/ARC, or equal. The indicator readout shall duplicate that of the driving source as directed by the interface wiring. Specific readouts for the source controlled mode of operation are:

a. Manual - The five or six digit readout of the channel frequency shall be presented.

b. Preset - The readout shall be the preset channel number selected. The remaining digits shall not be energized.

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c. Guard - The guard frequency on which the system shall be operating and as shown at the driving source shall be presented.

d. 243 - The frequency 243.000 shall be presented.

e. Preset read - The frequency of the preset channel in use shall be presented.

f. Test - Upon placing the system in test mode the display digits shall become blanked. Upon completion of the test cycle, all segments of the readout shall be illuminated (all 8's) or the fault location code shall be displayed, duplicating the source readout.

3.6.5.1 Special mode presentation. For the above conditions the display inhibit line at the main connector shall be open circuited. When the display inhibit line J1 pin 11 is grounded, the first digit readout shall be letter "A". The remaining digits shall present the frequency information duplicating that of the driving source readout.

3.6.6 Electrical connections. Connections to external circuits shall be provided on receptacles as specified in Table V.

Table V. Connections to external circuits

Reference Designation	Receptacle Type	Function
J1	Bendix TVPO 2RW-11-35P (MIL-C-38999, Series III)	Power/Control

3.6.6.1 Parallel control lines. The indicator shall interface with other equipment in the system using the wiring to J-1 as specified in Table VI.

3.6.6.2 Serial data bus. The indicator shall derive the readout information from the serial data bus system as specified in MIL-C-85666. The serial data bus system shall consist of a pair of wires for data and a pair of wires for the clock pulses. The characteristics of the lines are specified in Table VI (connector J1 pins 7, 8, 9 and 10).

CAUTIONARY NOTE
ARC-182 SERIAL CONTROL INTERFACE SPECIFICATION

All devices connected to the serial data bus should read and decode all data information. Further, the device should decode command word information to compare data sent versus data responses, to insure that no inconsistencies have occurred, either as a result of false clock pulses, bit transistions or external interference.

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Table VI. Indicator interface characteristics J1 - power/control connector

Pin No.	Pin Function	I/O	Characteristics
1	Panel Lamp	I	5 Vac or dc, 5 watts max; use either pin 1 or pin 2, not both
2	Panel Lamp	I	28 Vac or dc, 5 watts max; use either pin 2 or pin 1, not both
3	Panel Lamp	I	Power return for panel lamp power
4	Primary Power	I	System ground
5	On/Off	I	0 to +7 Vdc turns unit on; open circuit turns unit off
6	28 Vdc Power In	O	28 Vdc power per MIL-STD-704, 15 watts maximum
7	Data (High)	I	"1"=4.2 to 5.3 Vdc greater than 10K ohm load impedance. "0"=0 Vdc ± 0.5 Vdc
8	Data (Low)	I	See pin 7. Line is at "1" when not active.
9	Clock (High)	I	See pin 7. Line is at "0" when not active.
10	Clock (Low)	I	See pin 7. Line is at "1" when not active.
11	Display Inhibit	I	Open circuit for normal operation. 0 to ± 4 Vdc changes display as specified in 3.5.7.1
12	Spare		
13	Spare		

3.7 Safety. The indicator safety requirements shall be in accordance with MIL-E-5400 except resistance from ground to chassis shall be not greater than 20 milliohms.

3.7.1 Human performance/human engineering. MIL-STD-1472 and MIL-H-46855 shall be used as a design guide for the indicator and all component parts therein.

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3.8 Bonding. Bonding shall be in accordance with MIL-B-5087.

3.9 Workmanship. Workmanship for the indicator shall conform with MIL-STD-454, Requirement 9.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1. Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

4.3 Standard conditions. Unless otherwise specified herein, all inspections shall be performed with conditions specified as follows:

a.	Temperature	Room ambient (25 ± 10°C)
b.	Altitude	Normal ground
c.	Vibration	None
d.	Humidity	Room ambient up to 90 percent relative humidity
e.	Input Power	28 ± 0.5 volt dc

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4.4 First article inspection. First article inspection shall be conducted by the contractor on an indicator representative of the production indicators to be supplied under the contract. First article tests shall be accomplished under the authorized test procedures (see 4.6). No first article inspections shall be conducted prior to acceptance of the first article test procedure by the procuring activity. First article inspection shall consist of all the tests described in 4.4.1.

4.4.1 Scope of first article tests. First article tests shall include all tests in the approved test procedures to determine that the indicator meets all the requirements of this specification and the contract. First article tests shall include environmental tests in accordance with the procedures of MIL-T-5422 including temperature shock, a random vibration test as specified in 4.4.4, and an electromagnetic compatibility test in accordance with MIL-STD-461 and MIL-STD-462 as modified in 4.4.3. The first article test may also include a reliability qualification test.

4.4.2 First article approval. Approval of the first article sample will be by the procuring activity upon completion of all tests.

4.4.3 Electromagnetic interference tests. Compliance with the requirements of 3.4.4.3 shall be demonstrated by tests. These tests shall be performed in accordance with MIL-STD-462, with Notice 2 as modified herein.

4.4.3.1 EMI test plan. The contractor shall perform EMI tests in accordance with a contractor prepared EMI Test Plan. The Test Plan shall describe all EMI tests required to demonstrate compliance with the requirements of 3.4.4.3, and shall address all anomalies, inconsistencies, and omissions (if any). The Test Plan shall detail a combination of operating modes and frequencies to be tested. The combination shall represent all operating modes of the indicator and at least four frequencies. The Test Plan shall be acceptable to the procuring activity prior to the commencement of EMI testing.

4.4.3.2 Ground plane interference. The contractor shall prepare a procedure to demonstrate that the indicator complies with the requirements of 3.4.4.3.4. The procedure shall be included in the EMI Test Plan.

4.4.3.3 Conducted susceptibility test, interconnecting and signal leads. In addition to the requirements of MIL-STD-461 the following RF susceptibility test shall be conducted. No malfunction, undesirable response, or change in indication beyond the tolerance given in the equipment specification or approval test plan shall be produced in any equipment when the RF signal levels listed in Table VII are coupled into the interconnecting wiring.

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Table VII. Susceptibility test signals

Frequency	Required Signal Levels In One Turn Loop
2 to 30 MHz	100 MV RMS

The RF signal shall be modulated 80 percent with 400 Hz and with that frequency or type of modulation to which the test sample is most susceptible. This requirement applies to all interconnecting wires or cable between component parts of subsystems, and to wires and cables connected to other equipment. The RF signal shall be measured in the one-turn loop with an RF voltmeter as shown on Figure 2. The signal generator shall have an output impedance of 50 ohms or less and a maximum open-circuit voltage of not less than 3 volts rms. The test procedure shall be as follows:

a. With the current probe clamped around the wire bundles under test, set the signal generator for maximum output and slowly scan from 2 MHz to 30 MHz.

b. At those frequencies where a response is noted, move the current probe along the wire for maximum indication. Reduce the signal generator output and determine the threshold of susceptibility of the wire bundle. When a response is found on an interconnecting wire bundle, probe each wire in the bundle separately to determine the susceptible wire(s).

c. Record and include in the test report the threshold of susceptibility data (frequency, amplitude, and description of responses) obtained in (b) above. Also include in the test report the corrective action required for each wire that does not meet the required level.

Wire bundles may be tested in sections if the current probe will not fit around the complete wire bundles. Chassis or circuit grounds not normally routed with the interconnecting wiring shall be measured individually and shall not be included in the wire bundle under test. Power ground return leads shall not be placed in the probe.

4.4.4 Random vibration test. A random vibration test shall be conducted in accordance with the procedures of MIL-STD-810 and as specified in 4.4.4.1 through 4.4.4.4.

4.4.4.1 Test item operation. The test item shall be operated during application of random vibration so that the functional effects caused by these tests may be evaluated. The test item shall meet performance requirements, as specified, while the functional vibration levels are being applied and immediately preceding and following the application of the endurance levels.

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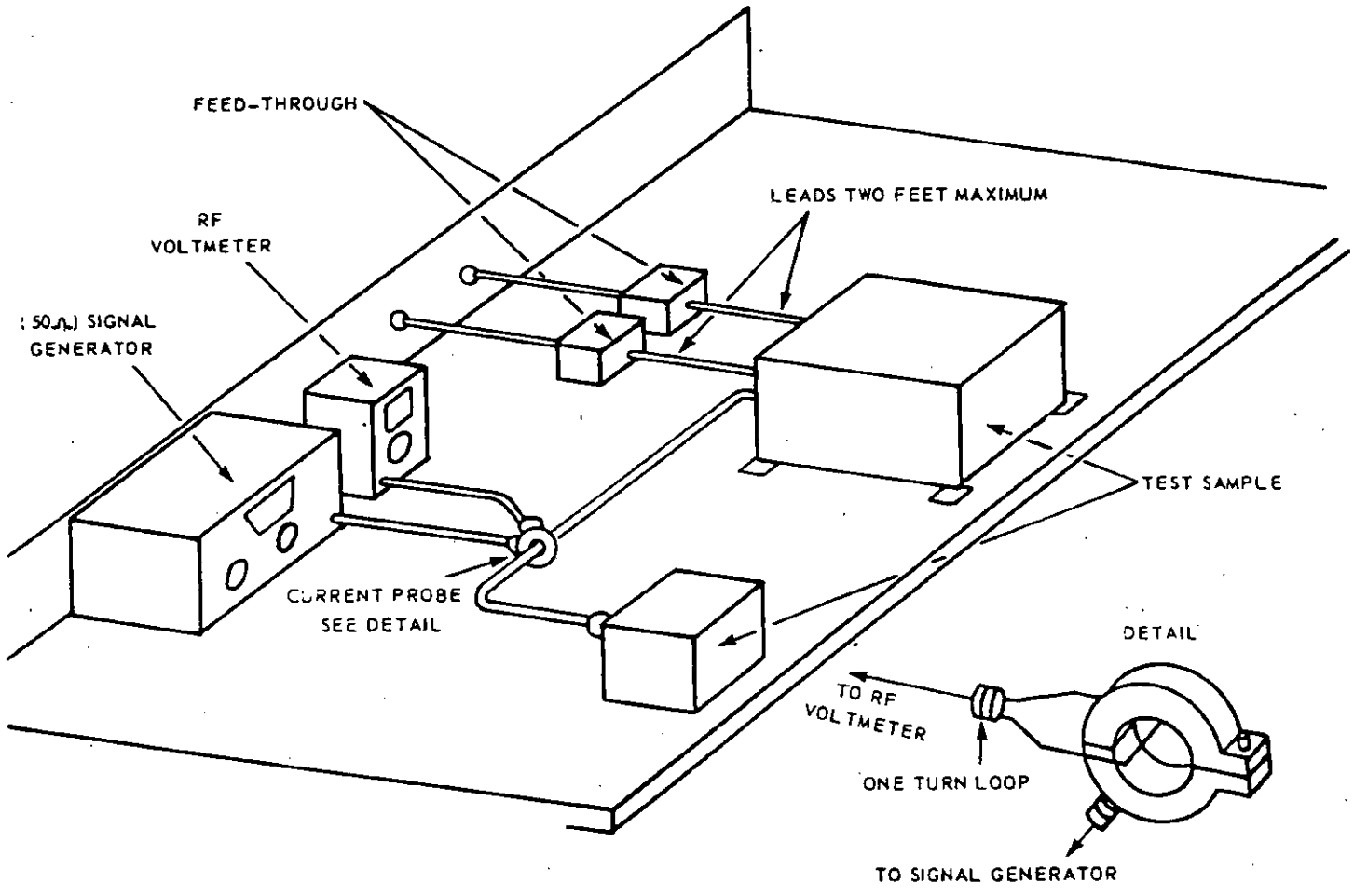


FIGURE 2. Conducted susceptibility test set up

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4.4.4.2 Mounting techniques. The test item shall be hard mounted either directly to the vibration exciter or transition table, or by means of a rigid fixture capable of transmitting the vibration conditions specified herein. Precautions shall be taken in the establishment of mechanical interfaces to minimize the introduction of extraneous responses in the test setup. The test load shall be distributed as uniformly as possible on the vibration exciter table in order to minimize effects of unbalanced loads. The input control sensing device(s) shall be rigidly attached to the vibration table, or fixture if used, as near as possible to the attachment point(s) of the test item. Additional vibration sensors shall be located in or on the test item to determine resonant frequencies and amplification factors. Locations to be selected should include main structure, printed circuit boards, large components, and modules, where practicable. The sensor sizes and weights shall be limited so that their effect on the dynamic responses being measured is minimal.

4.4.4.3 Performance of test. The individual equipment test item shall be subjected to broadband random vibration excitation with the power spectral density envelope shown on Figure 3. The test item shall be attached to the vibration exciter according to 4.4.4.2. Vibration shall be applied sequentially along each of the three orthogonal axes of the test item. Two test levels are required, a functional level and an endurance level. For each axis, one half of the functional test shall be conducted first, then the endurance test, followed by the second half of the functional test. The equipment shall perform according to the operating requirements as specified in 4.4.4.1. The acceleration power spectral density (g^2/Hz) of applied vibration, as measured on the test fixture at mounting points of the test item, shall be as specified in 4.4.4.4. Test times shall, for each axis, be one hour each for functional and endurance levels. The instantaneous random vibration acceleration peaks may be limited to three times the rms acceleration level. The power spectral density of the test control signal shall not deviate from the specified requirements by more than +100, -30 percent (+3, -1.5 dB) below 500 Hz and +100, -50 percent ($\pm$ 3 dB) between 500 Hz and 2,000 Hz except that deviations as large as +300, -75 percent ($\pm$dB) shall be allowed over a cumulative bandwidth of 100 Hz maximum, between 500 and 2,000 Hz. Tolerance levels in terms of dB are specified as:

$$dB = 10 \log_{10} \frac{W1}{W0}$$

Where W1 = measured acceleration power spectral density in g^2/Hz units, the term W0 defines the specified level in g^2/Hz units. Confirmation of these tolerances shall be made by use of an analysis system providing statistical accuracies corresponding to a bandwidth-time constant product, BT=50 minimum. Specific analyzer characteristics shall be as specified below or equal, subject to the BT=50, minimum limitation.

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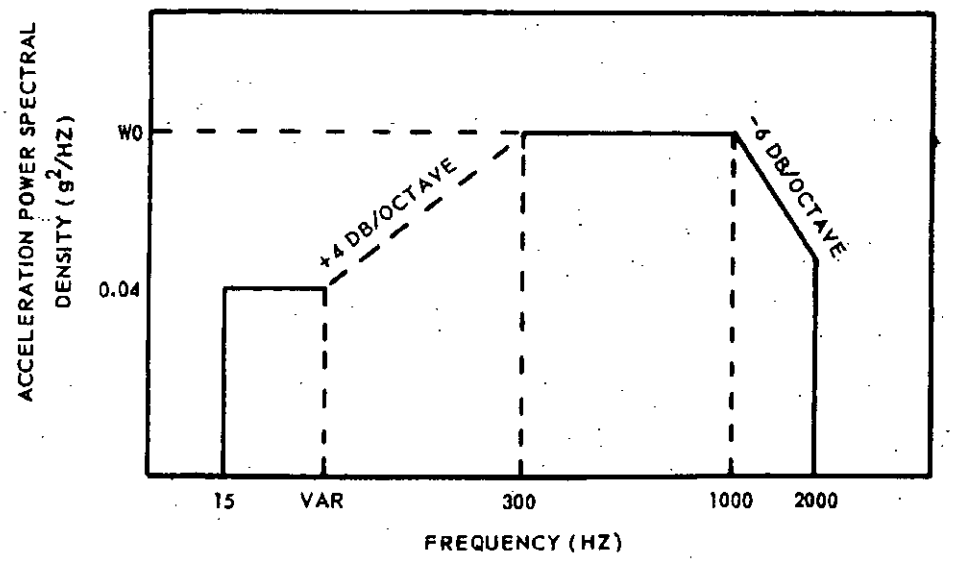


Figure 3. Random vibration envelope

a. On-line, contiguous filter, equalization analysis system having bandwidth as follows:

B=25Hz, maximum between 20 and 200 Hz
B=50 Hz, maximum between 200 and 1,000 Hz
B=100 Hz, maximum between 1,000 and 2,000 Hz

b. Swept frequency analysis system characterized as follows:

(1) Constant bandwidth analyzer.

(a) Filter bandwidth as follows:

B=25Hz, maximum between 20 and 200 Hz
B=50 Hz, maximum between 200 and 1,000 Hz

(b) Analyzer averaging time =T= 2 RC = 1 second, where T=true averaging time constant

(c) Analysis sweep rate (linear) =

$$R = \frac{B}{4RC} \text{ or } \frac{B^2}{8} \text{ y(Hz/second)}$$

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(2) Constant percentage bandwidth analyzer

(a) Filter bandwidth - pfc=one-tenth of center frequency maximum (0.1 fc) where p=percentage and fc=analyzer center frequency

(b) Analyzer average time = T = $\frac{50}{pfc}$ minimum

(c) Analysis sweep rate (logarithmic)

$$T = \frac{pfc}{4RC} \text{ or } \frac{(pfc)^2}{8}$$

(Hz/second), maximum, whichever is smaller

Digital power spectral density analysis employing quantization techniques providing accuracies corresponding to the above approach.

4.4.4.3.1 Worst case axis. During the vibration testing, data shall be obtained to identify the worst case axis for purpose of vibration screening tests on production equipments.

4.4.4.4 Test levels. For the purpose of the random vibration test, the following test-level shall be used:

a. Functional Level-W_o = 0.05 g^2/Hz (8.5g rms minimum)

b. Endurance Level-W_o = 0.12 g^2/Hz (12.7 g rms minimum)

4.4.5 Reliability qualification test. A reliability qualification test may be included in the first article tests. A minimum of six indicators shall be tested in accordance with MIL-STD-781 using the reliability qualification test - mission profile in this specification. (See Figures 4 and 5.) Each indicator shall have successfully completed the normal burn-in and all equipment test (AET) prior to the start of RQT. The mission profile test cycle should begin at the "12 hour" point which is a cold day ground stabilization (-54°C) prior to ground operation and flight in a cold day. Random vibration should be applied with the units hard-mounted (without vibration isolators). The accept-reject criteria imposed shall be those of MIL-STD-781, Test Plan IIIC, with the upper test MTBF specified in 3.4.1.3, except that the test shall be truncated after 10,300 hours.

4.4.6 Maintainability demonstration. A maintainability demonstration for intermediate level requirements shall be conducted in accordance with MIL-STD-2084 and MIL-STD-471. Test Method 1, Test Plan B of MIL-STD-471, shall be used.

4.4.7 Production indicators. Indicators supplied under the contract shall in all respects, including design, construction, workmanship, performance and quality, be capable of successfully passing the same tests as specified herein.

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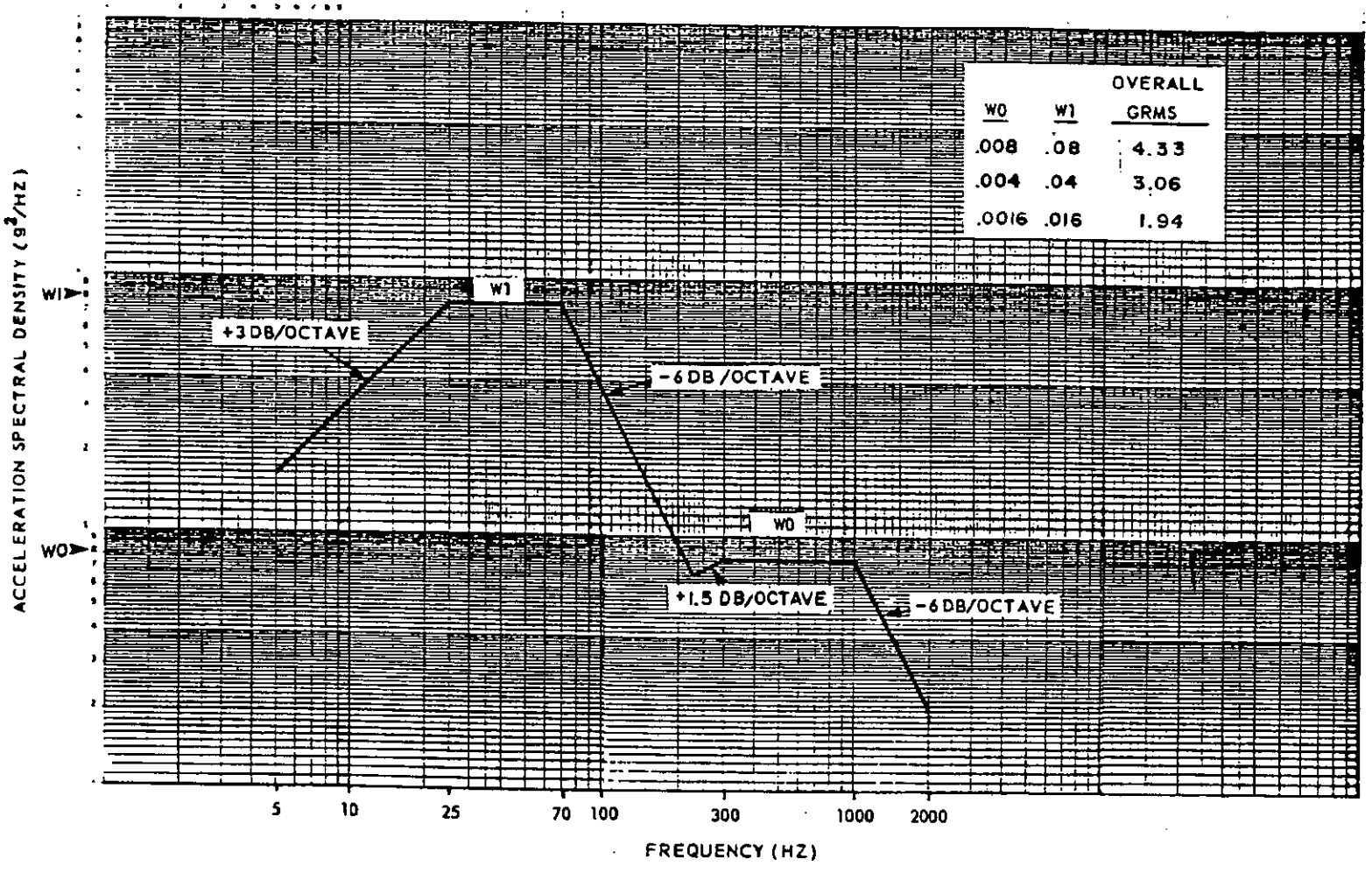


FIGURE 4. RQT vibration envelope

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NOTES:

1. THE CHAMBER AIR TEMPERATURE RATE OF CHANGE SHALL BE 5° C/MINUTE
2. THE INPUT VOLTAGE SHALL BE TURNED OFF LONG ENOUGH TO STABILIZE
3. DURING THE TEST THE INPUT VOLTAGE SHALL BE ADJUSTED BETWEEN 30.8 VDC 28.0 VDC AND 25.2 VDC THE EQUIPMENT SHALL BE OPERATED FOR 7.5 " ON" HOURS (ALTERNATELY) AT EACH INPUT LEVEL

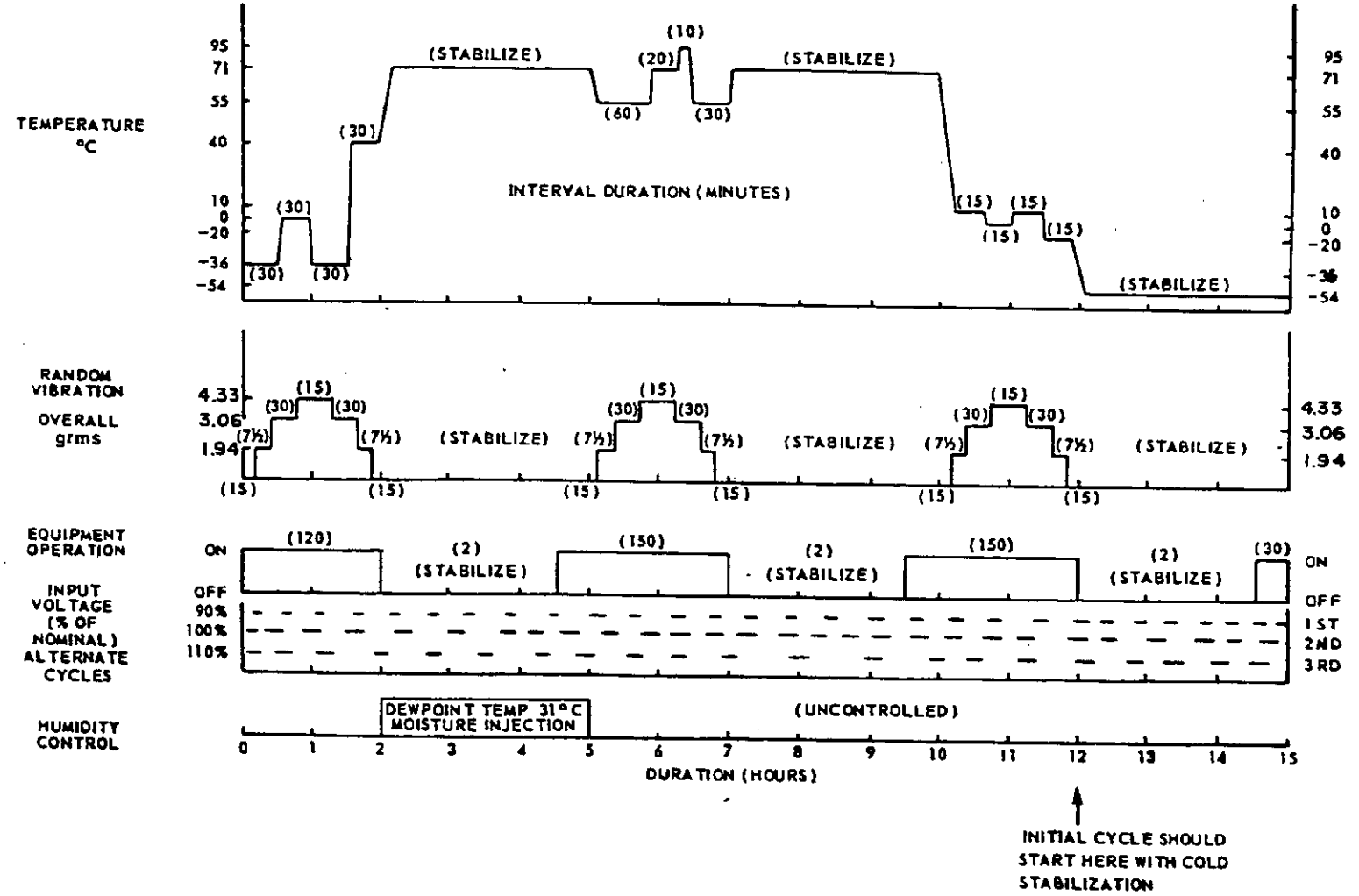


FIGURE 5. RQT mission profile test cycle

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4.5 Quality conformance inspection. The contractor shall furnish all samples and shall be responsible for accomplishing the quality conformance tests. All inspection and testing may be under the supervision of the customer representative. The contractor shall retain test data showing quantitative results for all quality conformance tests. Such tests shall be signed and stamped by an authorized representative of the contractor or laboratory, as applicable. Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product. Quality conformance tests shall consist of the following:

a. Individual tests (see 4.5.1)

b. Manufacturing screening (see 4.5.2)

c. Sampling tests (see 4.5.3)

d. Reliability assurance tests (see 4.5.4)

e. Special tests (see 4.5.5)

4.5.1 Individual tests. Each indicator submitted for acceptance shall be subjected to the individual tests. The tests shall be adequate to determine compliance with the requirements of materials, workmanship, operational adequacy and reliability. As a minimum, each indicator accepted shall have passed the following tests:

a. Examination of product (see 4.5.1.1)

b. Operational test (see 4.5.1.2)

4.5.1.1 Examination of product. Each indicator shall be examined carefully to determine that the material and workmanship requirements have been met.

4.5.1.2 Operational test. Each indicator shall be operated long enough to permit the equipment temperature to stabilize and to check sufficient characteristics and record data to assure specified equipment operation.

4.5.2 Manufacturing screening. The contractor shall subject each WRA to manufacturing screening prior to undergoing sample tests or reliability assurance tests. Manufacturing screening shall consist of WRA random vibration and temperature cycling. An additional screening period may be used at the option of the contractor. If an additional screening period is to be used, the details thereof shall be included in the approved test procedures and shall be the same for all equipment, including spare units.

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4.5.2.1 WRA random vibration. Prior to conducting WRA temperature cycling, each WRA shall have completed a 10-minute failure-free vibration screen at 6.1g rms. Power spectral density shall cover 20-2000 Hertz. Pseudo-random vibration methods are acceptable. Vibration envelope is shown on Figure 6. The test item shall be hard mounted without vibration isolators and shall be operating throughout the test.

4.5.2.2 WRA temperature cycling. Each WRA shall undergo 25 temperature cycles between -54°C and 71°C. The rate of temperature change between extremes shall be at least 5°C per minute. When the WRA temperature has stabilized within 5°C or either temperature extreme, the WRA shall be soaked. Hot soak at +71°C shall be two hours; cold soak at -54°C duration shall be one hour. At the conclusion of each soak, the next temperature ramp shall begin. The WRA shall be energized throughout temperature cycling except during the cooldown period from +71°C to -54°C and during the 60 minute soak at -54°C. In addition, the WRA shall be turned off for one minute at points 30, 60, and 90 minutes into each hot soak period. Indicator performance shall be periodically checked during each cycle and repairs shall be made to correct failures. The last five consecutive temperature cycles shall be failure free. One temperature cycle is defined as: operation at +71°C for the 120 minute hot soak (including three minutes "off"), non-operational cooldown to -54°C, 60 minute non-operational cold soak at -54°C, operational temperature increase to +71°C.

4.5.3 Sampling tests. Indicators shall be subjected to sampling tests. Indicators selected for sampling tests shall first have passed the individual tests. Indicators will be selected for sampling tests by the government inspector in accordance with Table VIII.

Table VIII. Sampling test sample selection

Quantity of equipments offered for acceptance	Quantity to be selected for sampling test
First 10	1
Next 50	1
Next 75	1
Next 100	1
	1 for each additional 200 or fraction thereof

Sampling tests are not required when reliability assurance tests are conducted.

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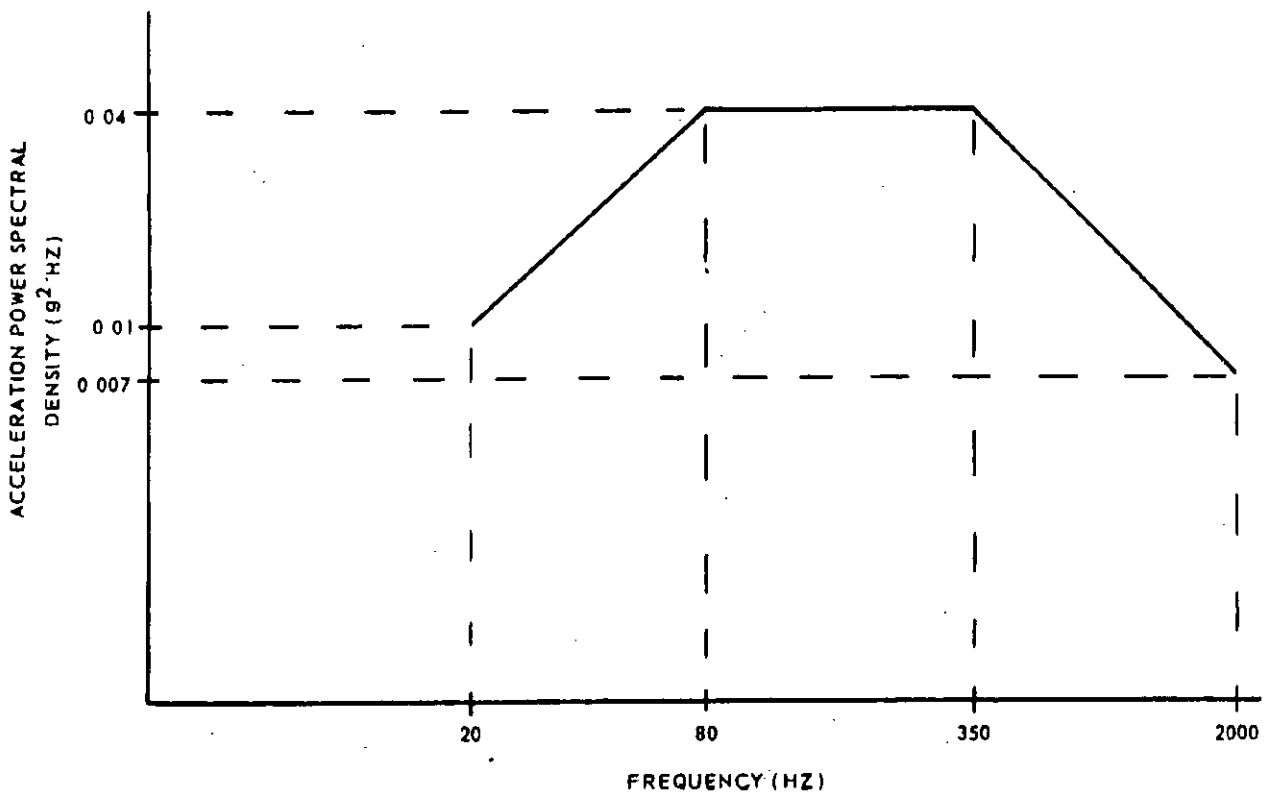


FIGURE 6. Manufacturing screening random vibration envelope (see 4.5.2.1)

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4.5.3.1 Scope of sampling tests. As a minimum, each indicator selected for sampling tests shall be subjected to the following tests:

a. Complete operational test at ambient room conditions, making all necessary measurements to assure that all applicable specification requirements have been met.

b. Operational test at certain environmental conditions. The conditions may vary for each indicator tested and should be based on results of the first article, individual and special tests.

c. During the reliability sample test specified in 4.5.4.2, each failure shall be analyzed as to cause and remedial action necessary to reduce the possibility of its recurring in future equipment taken.

4.5.4 Reliability assurance tests. A reliability assurance test shall be conducted using MIL-STD 781. Classification of failure shall be in accordance with 4.5.7.

4.5.4.1 All equipment test. Each indicator shall be tested as specified in MIL-STD-781 and modified herein. Temperature cycling and power on/off cycling shall be as shown on Figure 5. No vibration is required. Each indicator shall have successfully completed manufacturing screening as specified in 4.5.2 prior to the start of the test. Each equipment (WRA) shall be tested separately as opposed to testing as an equipment system. The test MTBF shall be as listed in 3.4.1.3. To determine whether the MTBF is being met at any time during the contract, the operating time and the failures therein (not counting any burn-in failures or burn-in operating time) shall be totaled and the results compared with the reject line of Test Plan XVIIIC of MIL-STD-781. (Extend the line as necessary to accommodate the data.) These totals shall accumulate so that at any time the experience from the beginning of the contract is included. At any time that the current totals of test hours and test failures plotted on Test Plan XVIIIC curves show a reject situation, the procuring activity shall be notified. The procuring activity reserves the right to stop the acceptance of indicators at any time that a reject situation exists pending a review of the contractors efforts to improve the equipment, the equipment parts, or the equipment workmanship so that the entire compilation will show other than a reject decision.

4.5.4.2 Reliability sample test. When specified in the contract or purchase order (see 6.2.1.), a monthly sample test of equipments shall replace the reliability test of 4.5.4.1. Units shall be tested as specified in MIL-STD-781 as modified herein. Temperature cycling and power on/off cycling are required and shall be as shown on Figure 5. Each sample equipment (WRA) shall be tested for 256 hours with the last two cycles failure free. For determining MTBF compliance each designated equipment (WRA) shall be tested separately as opposed to testing as an equipment system. The MTBF for each designated equipment shall be those listed in 3.4.1.3. To determine whether the MTBF is being met at any time during the contract the operating test hours

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and the failures therein (not counting any burn-in failures or burn-in operating time) shall be totaled and the results compared with the reject line of Test Plan IIIC of MIL-STD-781. (Extend the line as necessary to accommodate the data.) These totals shall accumulate so that at any one time the experience from the beginning of the contract is included. The sample size shall be based on the production quantity as specified in Table IX.

Table IX. Reliability test sample size

Production Quantity	Sample Size
2 - 5	1
6 - 25	2
26 - 50	3
51 - 100	4

4.5.4.3 Test details. Test details such as the performance criteria to be measured, special failure criteria, and preventive maintenance to be allowed during the test shall be part of the test procedures to be prepared by the contractor prior to the beginning of the reliability assurance tests.

4.5.5 Special tests. When specified in the contract, special tests shall be conducted for the purpose of checking the effect of any design or material change on the performance of the indicator and to assure quality control (see 6.2.1). The indicator selected for special tests may be selected from the indicators previously subjected to the sampling or reliability tests.

4.5.5.1 Special test schedule. Selection of indicators for special tests shall be made as follows:

a. On an early equipment after an engineering or material change.

b. Whenever failure reports or other information indicate that additional tests are required. (This will be determined by the procuring activity.)

4.5.5.2 Scope of special tests. Special tests shall consist of such tests as authorized by the procuring activity. Test procedures previously approved for the first article tests shall be used where applicable. When not applicable, the contractor shall prepare a test procedure for the procuring activity acceptance prior to conducting the tests.

4.5.6 Equipment failure. Should a failure occur during either the sampling, reliability qualification, reliability assurance or special tests, the following action shall be taken:

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a. Determine the cause of failure and the equipment BIT performance in detecting and isolating the failure.

b. Determine if the failure is an isolated case or design defect or pattern failure.

c. Prepare proposed corrective action intended to reduce the possibility of the same component or BIT failure(s) occurring in future tests or documented justification where no corrective action is proposed.

d. Where practical, include a test in the individual test to check all indicators for this requirement until assurance is obtained that the defect has been corrected.

4.5.7 Failure classification. All incidents occurring during reliability assurance tests shall be classified and reported as either relevant or non-relevant in accordance with MIL-STD-2074 and MIL-STD-781. Only those incidents classified as relevant failures shall be accountable for making an accept/reject decision.

4.5.7.1 Relevant failures. All test incidents shall be considered as relevant failures unless determined to be non-relevant in accordance with the criteria of MIL-STD-2074.

4.5.7.2 Non-relevant failures. Although non-relevant failures are not used for reliability assurance test accept/reject decisions, all test article incidents and equipment failures shall be recorded and reported.

4.6 Test procedures. The procedures used for conducting all tests required by Section 4 shall be prepared by the contractor for review and acceptance. The right is reserved by the procuring activity or the government representative to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of the specification or the contract (see 6.2.1). MIL-T-18303 shall be used as a guide for preparation of test procedures.

4.7 Testing of spares and repair parts. Equipment delivered as spare or repair parts shall undergo acceptance tests to ensure that the items meet performance and reliability requirements not less than those imposed in production.

4.7.1 Weapons replaceable assemblies. All items to be delivered as spares at the weapons replaceable assembly (WRA) or higher level shall receive performance, manufacturing screening and reliability assurance tests commensurate with that afforded the original production equipment.

4.7.2 Shop replaceable assemblies. All items to be delivered as spares at the shop replaceable assembly (SRA) or sub-SRA level shall receive functional checks and screening under conditions and for a duration commensurate with that imposed on that assembly in the original production.

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4.7.3 Repair parts. All electronic component parts to be delivered as repair parts shall receive functional and screening tests equal to those imposed for the nearest military standard part.

4.7.4 Test details. The details of acceptance and screening tests to be performed on SRA's and WRA's supplied as spare or repair parts shall be incorporated into item specifications, drawings and test procedures.

4.8 Presubmission testing. No item, part, or complete equipment shall be submitted until it has been previously tested by the manufacturer and found to comply with all applicable requirements of this specification and the contract.

5. PACKAGING

5.1 General. All major units and parts of the indicator shall be preserved, packaged, packed, and marked in accordance with MIL-E-17555 for the level of shipment specified in the contract or purchase order (see 6.2.1).

5.2 Marking. Shipments shall be marked in accordance with requirements of MIL-STD-129.

6. NOTES

6.1 Intended use. The indicators specified herein are one portion of the AN/ARC-182(V) radio system. The AN/ARC-182(V) is a variable configuration of units using common modules that shall be used to meet the prime and backup UHF voice plus prime UHF/VHF/AM/FM voice and data commmunication requirements in any tactical aircraft. The variable configurations are intended to provide direct physical replacement of AN/ARC-159(V) radios. They are intended for production incorporation in new aircraft plus retrofit in selected aircraft.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification

b. If a first article inspection is required (see 3.2, 6.4)

c. If non-standard parts or materials are required (see 3.3.5)

d. If reliability program is not to be established by contractor (see 3.4.1)

e. Whether All Equipment Test or Reliability Sample Test is required (see 4.5.4.1, 4.5.4.2)

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f. If Special Tests are required (see 4.5.5).

g. Test procedure modifications or additional tests, if required (see 4.6).

h. Selection of applicable levels of packaging and packing (see 5.1).

6.2.2 Data requirements. When this specification is used in an acquisition and data are required to be delivered, the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (CDRL), incorporated into the contract. When the provisions of DOD FAR Supplement, Part 27, Sub-Part 27.410-6 (DD Form 1423) are invoked, and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs:

Paragraph	Data requirement title	Applicable DID no.
3.3.2	Engineering Change Proposals (ECPs) and Requests for Deviations and Waivers (Long Form)	DI-E-2037
3.3.3	Program Parts Selection List (PPSL)	DI-MISC-80072
3.4.1.2	Part Approval Request	DI-MISC-80071
3.4.8	Request for Confirmation of Nomenclature	DI-CMAN-80194
	Request for Assignment of Serial Number and Serial Number Prefix Letters	DI-CMAN-80195
	Request, Assignment of Nomenclature	UDI-E-21582
	Requests for Approval of Identification Plate Drawings	DI-CMAN-80196
4.4, 4.6	First Article Inspection Procedure	DI-T-4901
4.4.2	First Article Inspection Report	DI-T-4902
4.4.3.1, 4.4.3.2, 4.4.3.5	Electromagnetic Interference Test Plan	DI-EMCS-80201
4.4.3.3	Electromagnetic Interference Test Report	DI-EMCS-80200

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Paragraph	Data requirement title	Applicable DID no.
4.5.4.1, 4.5.4.2	Reliability Test Report	DI-RELI-80252
4.5.4.3	Reliability Test Procedures	DI-RELI-80251
4.5.5.2	Production/Acceptance Inspection Procedures	DI-T-4903
4.5.6c	Plan, Failure Data Collection, Analysis and Corrective Action	UDI-T-23719
4.5.7.2	Failed Item Analysis Report	DI-RELI-80253
4.7.4	Drawings, Engineering and Associated Lists	DI-E-7031

(Data item descriptions related to this specification, and identified in section 6 will be approved and listed a such in DoD 5010.12L, AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.3 Options. The following options are available:

a. An additional manufacturing screening period is allowable at the option of the contractor (see 4.5.2).

b. Acceptance of equipment may be stopped anytime a reject situation exists in the All Equipment Test. This is at the option of the procuring activity (see 4.5.4.2).

6.4 First article. When a first article inspection is required, the item should be a first article sample. The first article should consist of one unit. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, may furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.5 Associated equipment. The indicator should operate with the associated equipment listed in Table X which should not be supplied as part of this equipment:

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Table X. Associated equipment

Item	Equipment Designation	Military Specification
ANVIS Goggles	AN/AVS-6	ADS-23
Receiver-Transmitter, Radio	RT-1324A/ARC	MIL-R-85665
	RT-1327A/ARC	MIL-R-85665
	RT-1407/ARC	MIL-R-85665
Control, Switching Unit	C-11628/ARC	MIL-C-84674
	C-10777A/ARC	MIL-C-84674
	C-11132/ARC	MIL-C-84674
Control, Radio Set	C-11128/ARC	MIL-C-85806
	C-11129/ARC	MIL-C-85806
	C-11130/ARC	MIL-C-85806
Receiver-Transmitter	RT-1250A/ARC	MIL-R-85664

6.6 Subject term (key word) listing.

AN/ARC-182(V) Radio Set
Built-In-Test (BIT)
Blue Filtered White Panel Lighting
Frequency Channel
Indicator
Red Panel Lighting
Remote Mounted
Ultra High Frequency (UHF)
Very High Frequency (VHF)
White Panel Lighting

Preparing Activity
Navy AS

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